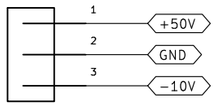
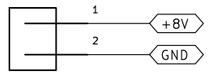
CLASS B / OPA810 + VAS — interfaces and assembly map

Frozen baseline, 21 Oct 2026. 2 op amps / 11 BJT. Component values unchanged.

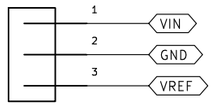
J1
HV SUPPLY



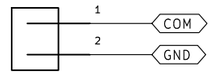
J2
LV SUPPLY

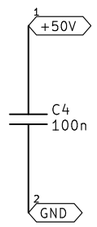


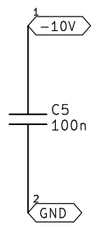
J3
VIN + REFERENCE

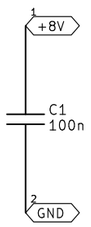


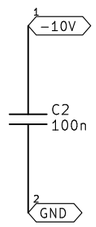
J4
COM OUTPUT



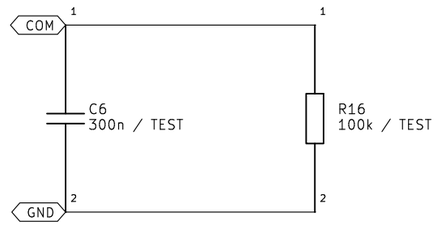








VIN: 0...2.5 V
VREF: 165.4846 mV (external)
COM target: 17.92 VIN − 2.8 V
Op amps: +8 / −10 V ONLY
Power stage: +50 / −10 V



Open the four functional sheets below. Wires join at dots; crossing without dot is not connected.
OUT_INT is before L1. HF feedback comes from DRIVE, not VAS_OUT.
Bench prototype: ramp ringing remains; current clamps are not SOA shutdown.
Bulk rail capacitors and a +8 V regulator are not included in this frozen simulation.

01 — Error amplifier / feedback

File: control.kicad_sch

02 — Voltage gain / active load

File: vas.kicad_sch

03 — Diamond current driver

File: diamond.kicad_sch

04 — Class B power / limiters

File: power.kicad_sch

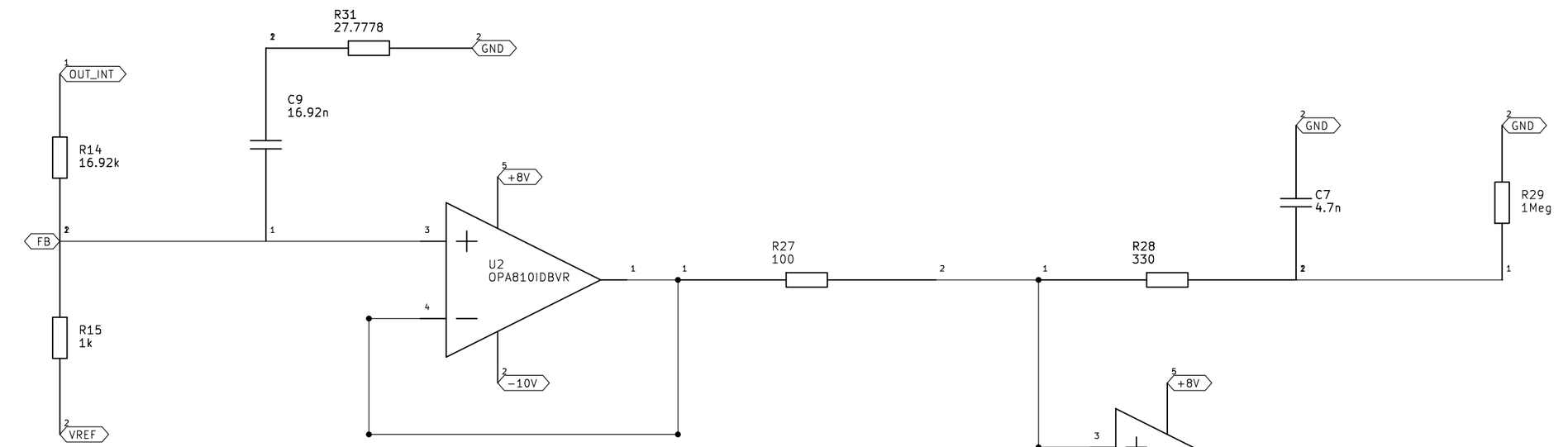
Native wired schematic; SPICE connectivity audited
Epson FA04010 driver research

Sheet: /
File: class_b_lv.kicad_sch

Title: CLASS B / OPA810 + VAS — interfaces and assembly map

Size: A3	Date: 2026-10-21	**Rev: LV-B baseline / review**
KiCad E.D.A. 9.0.7		Id: 1/5

01 — Low-voltage error amplifier and dual feedback

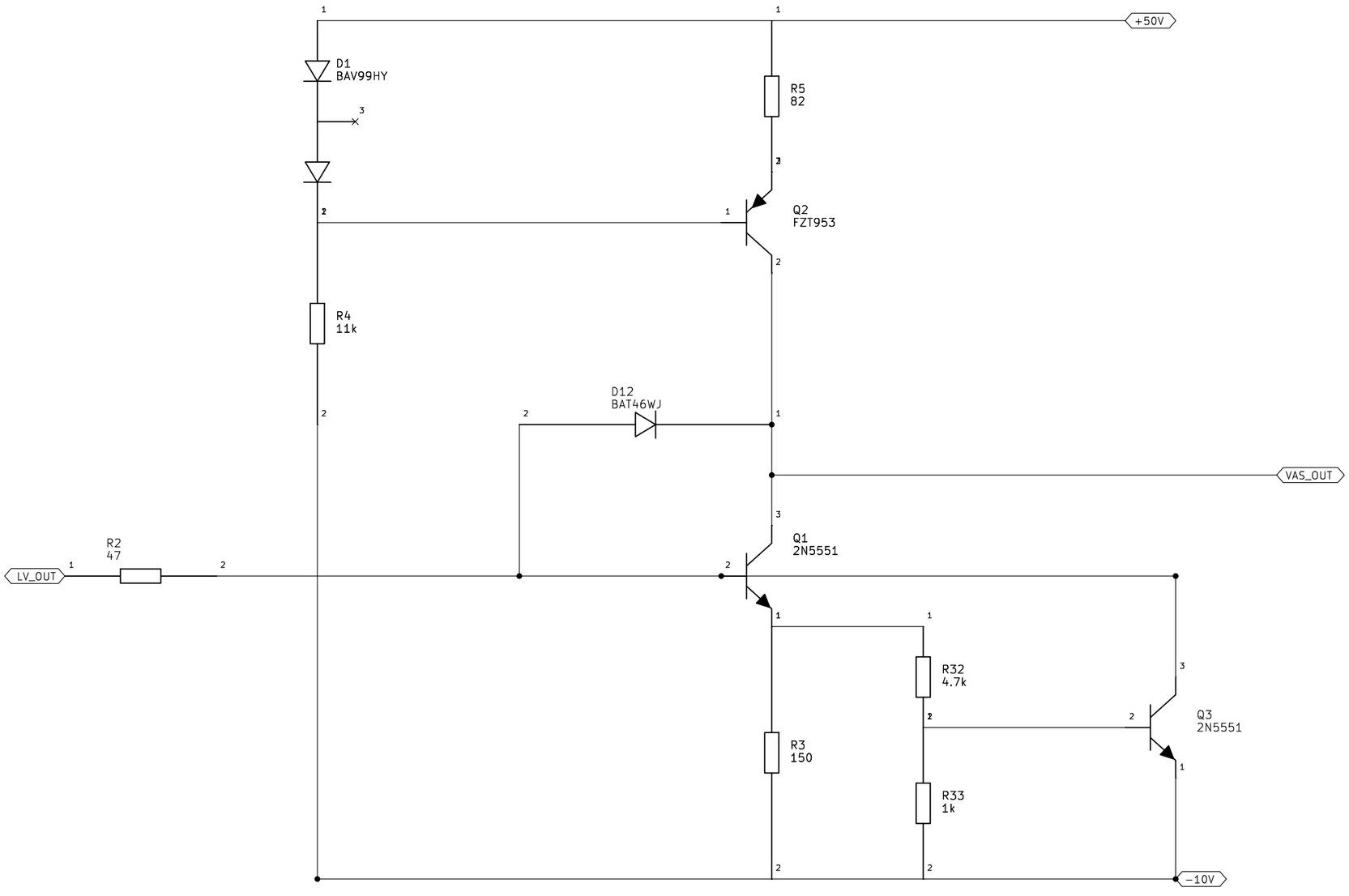


U1 polarity includes the inverting VAS on sheet 02.
Do not swap the op-amp inputs to match a conventional non-inverting amplifier.
CLOCAL / CDIFF and their resistors form matched frequency-dependent branches.
Values shown are exact simulation targets; tolerance and E-series rounding remain to be checked.

DBV / SOT-23-5:
1 OUT, 2 V−, 3 IN+, 4 IN−, 5 V+

Native wired schematic; SPICE connectivity audited
Epson FA04010 driver research
Sheet: /01 — Error amplifier / feedback/
File: control.kicad_sch
Title: 01 — Low-voltage error amplifier and dual feedback
Size: A3 | Date: 2026-10-21 | **Rev: LV-B baseline / review**
KiCad E.D.A. 9.0.7 | Id: 2/5

02 — Common-emitter VAS, active current load and clamps



Q1: voltage gain / level shift
Q2: active collector load
Q3: VAS current clamp
D12: Baker clamp

D1 contains two series junctions.
Their junction is physical pin 3.

Q1 / Q3: 2N5551
Central TO-92: 1 E, 2 B, 3 C

Q2: FZT953
SOT-223: 1 B, 2 C, 3 E
Tab is collector (pin 2).

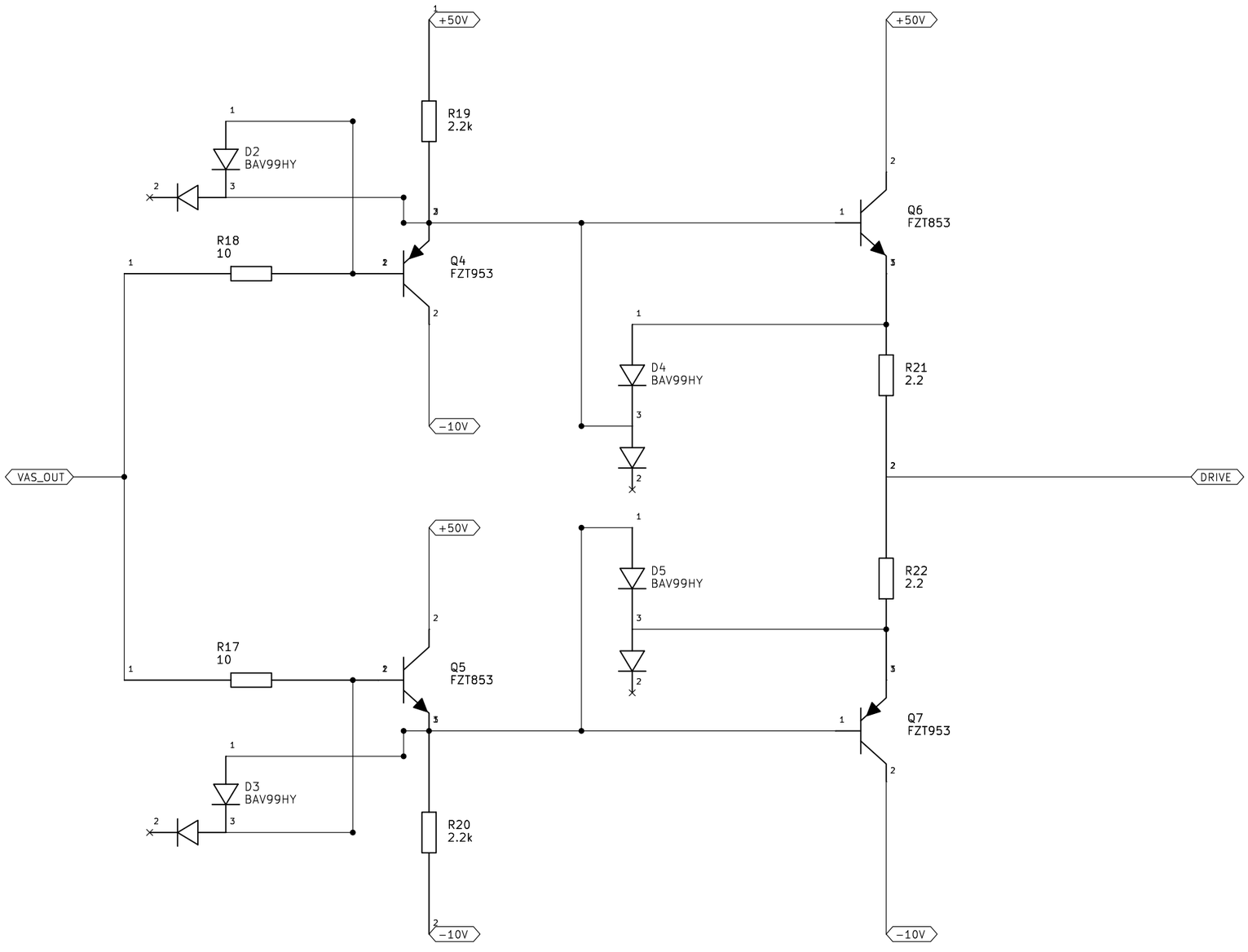
Native wired schematic; SPICE connectivity audited
Epson FA04010 driver research

Sheet: /02 — Voltage gain / active load/
File: vas.kicad_sch

Title: 02 — Common-emitter VAS, active current load and clamps

Size: A3	Date: 2026-10-21	**Rev: LV-B baseline / review**
KiCad E.D.A. 9.0.7		Id: 3/5

03 — Diamond current driver (four FZT transistors)



VAS_OUT is the voltage-stage output.
DRIVE is the current-buffer output.
They must not be shorted together.

Only this driver has standing bias.
The MJL power pair has no VBE spreader.

D2...D5 use one junction per package.
Second junctions are drawn locally:
pin 3 stays connected, pin 2 is open.
Cross = intentionally unconnected pin.

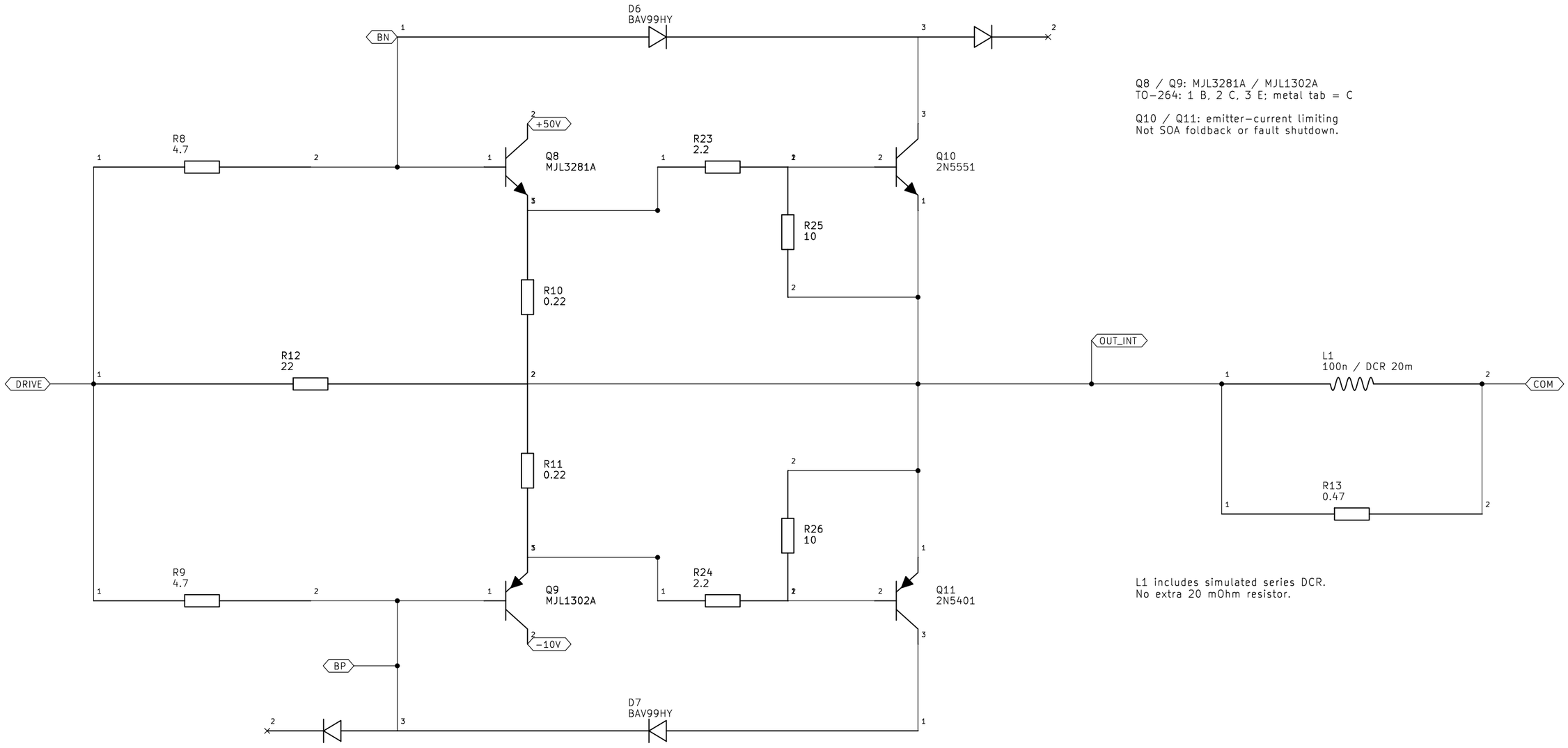
Native wired schematic; SPICE connectivity audited
Epson FA04010 driver research

Sheet: /03 — Diamond current driver/
File: diamond.kicad_sch

Title: 03 — Diamond current driver (four FZT transistors)

Size: A3	Date: 2026-10-21	**Rev: LV-B baseline / review**
KiCad E.D.A. 9.0.7		Id: 4/5

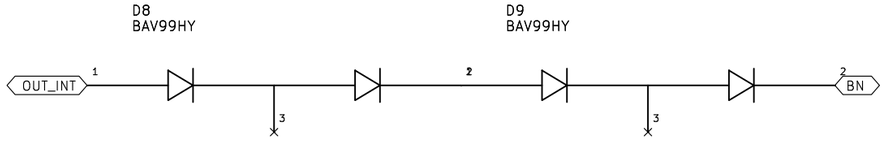
04 — Power emitter followers, current clamps and output isolation

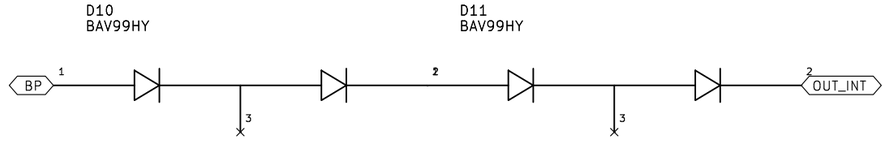


Q8 / Q9: MJL3281A / MJL1302A
TO-264: 1 B, 2 C, 3 E; metal tab = C

Q10 / Q11: emitter-current limiting
Not SOA foldback or fault shutdown.

L1 includes simulated series DCR.
No extra 20 mOhm resistor.





Native wired schematic; SPICE connectivity audited
Epson FA04010 driver research

Sheet: /04 — Class B power / limiters/
File: power.kicad_sch

Title: 04 — Power emitter followers, current clamps and output isolation

Size: A3	Date: 2026-10-21	**Rev: LV-B baseline / review**
KiCad E.D.A. 9.0.7		Id: 5/5